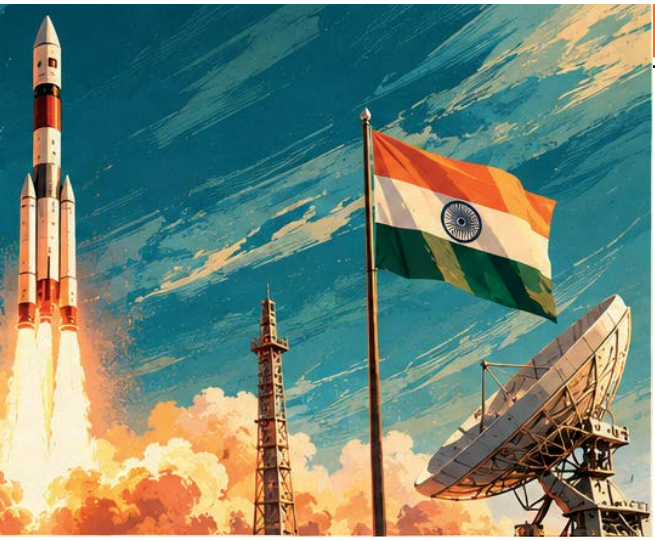


- Bellatrix abandoned launch vehicles for propulsion, while Digantara moved from debris tracking to defence, showing how companies are adapting as the market takes shape
- Reusable rockets have dramatically lowered launch costs, while SpaceX's scale in launches, Starlink and government contracts makes it difficult for smaller companies to compete head-on
- Geopolitical tensions are increasing the value of sovereign space capabilities, but Indian companies ultimately need repeat customers, large contracts and global markets to survive

THE SPACEX SHADOW

Total PE-VC capital raised by India's private space tech sector, 2019–2026 YTD

$1 billion

SpaceX Pentagon contracts, 2026 alone

$8 billion

India, private sector: commercial launches planned in FY27 and FY28

Eight

Space X launches in 2025

165

founder and CEO Awais Ahmed. "The first thing for any space company to do is get paying customers, so that you can sustain yourself through revenues and not from raising financing."

With funding scarce at the time, Pixxel had to earn its way on revenue, and the customers were abroad. "India has opened up, but (it is) still fairly early. The US and Europe have the numbers, much larger budgets for space, even in defence and intelligence," says Ahmed.

Going global forced a harder question: how does a small Indian player win against bigger, better-funded rivals? Pixxel's answer was to avoid competing on price at all.

"Our thinking was: where is the gap that we can fill where we don't have to compete on price?" says Ahmed. That gap was found in hyperspectral imaging, which captures data across narrow wavelengths of light, including ranges invisible to the human eye.

Where an ordinary satellite can only tell a farm from open ground and read the basic health of a crop, hyperspectral data can show the nutrients in the soil, its moisture levels, a crop's species, and the fertiliser it needs. It can also detect chemical, biological, and physical changes in the earth's surface that conventional optical satellites cannot.

US space agency NASA launched its first hyperspectral imaging satellite back in 2000, but it was decommissioned in 2017. Pixxel moved to fill that gap.

"We started providing a data set that shows things to people they can't see otherwise," Ahmed says. The bet paid off. "We ended up signing a contract with NASA and NRO (US National Reconnaissance Office). And that was the vindication. Build something so unique that people anywhere will pay a premium for it."

Finding customers, however, is only the first step. The harder part is generating enough repeat demand to build a market.

For most Indian companies, the value of space data remains difficult to translate into an immediate business case, say experts. Take agricultural lending. Satellite imagery can show what is happening to a crop, but the commercial prospect hinges on whether a